

# Team Science with Kendra

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## Team Science for Ecology

- Kendra Spence Cheruvilil (she | hers)
- Michigan State University, USA
- SPEC School, May 2026
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## Some talk norms

- Be present, as you can. Feel free to turn on your camera and ask questions as they come up.
- Take care of yourself (stand, leave, turn off camera, eat/drink, etc...).
- Be mindful of the space you take; be courageous in taking up space.
- If something is not working for you, propose an alternative way forward.

## A little about me...

## ...and my career path

### Part 1: Lessons learned doing big-data lake ecology

- Kendra Spence Cheruvilil & Patricia A. Soranno
- Co-directors of the Bridge Lab
- bridgelabresearch.org
- **Our idea:** To understand & predict lake water quality at the population level. But, first...

## Build the theory

- Landscape limnology
- Macrosystems ecology

## Build the research infrastructure

- Data & Tools
  - 1000's of ecosystems
  - 100's of drivers
  - multiple scales of space & time
- Lakes of the United States
  - LAGOS
  - lagoslakes.org

## Research infrastructure: The LAGOS-US Platform

- Facilitated by open science movement (100's of datasets)
- Grounded in open-science FAIR principles
- Curated databases, code, tools, and user guides
- Accessible, reproducible, and extensible

## Build the research infrastructure

- LAGOS-US Research Platform
  - GEO
  - LOCUS
    - \* RESERVOIR
      - Classification: natural lakes or reservoirs
    - \* DEPTH
      - Lake depths (maximum and minimum)
    - \* HUMAN
      - Human population, urbanization, classification, and lake access
    - \* NETWORKS
      - Lake network metrics of surface connectivity
    - \* LANDSAT
      - Predictions of CHL, Secchi, DOC, true color, TSS, and turbidity
  - LIMNO

## Research infrastructure: LAGOS research teams

- LAGOS Thru Time (2005-2025)
  - Increases in spatial resolution
  - Increases in spatial extent
  - Increases in number of lakes
  - Increases in people, disciplines
- Landscape Limno Research Group
  - 2005-2010
  - 6 states
  - ~2000 lakes  $\geq$  10 ha
  - Disciplinary: Aquatic ecologists

## LAGOS research teams

- **CSI Limnology**
  - 2011-2016
  - 17 states
  - ~50,000 lakes *ge* 4 ha
  - Interdisciplinary: Aquatic ecologists, GIS experts, Database experts, Data scientists (stats & comp science)
- **Continental Limnology**
  - 2016-2023
  - 48 states
  - ~500,000 lakes *ge* 1 ha
  - Interdisciplinary: MORE data scientists

## Science teams building and researching with LAGOS

- Build a diverse team of experts
- Best science outcomes for individuals and team

- Needs:
  - Trust
  - Fairness
  - Clarity
  - Transparency
  - Freedom
  - Understanding
- Challenges:
  - Power differentials
  - Unwritten Rules
  - Perceived constraints
  - Conflicts
  - Skills deficits
  - Disciplinary gaps

## Science of Team Science to build & foster productive research teams

- Team Diversity + Interpersonal skills
- Team functioning + Team communication
- Research outcomes
  - Generation/publication of transformative knowledge
  - Creation of new high-performing teams
  - Translation of sound management, conservation and policy
  - Innovative training and education
  - Effective public engagement

## How to build & foster productive research teams

- **Dimensions of team diversity:**
  - 1) Demographics
  - 2) Points of view
  - 3) Career stage
  - 4) Degree of incumbency
  - 5) Interaction modes
  - 6) Disciplinary balance
- **Interpersonal skills = learned & practiced**
  - Social Sensitivity
    - \* Empathy
    - \* Honesty
    - \* Clarity
    - \* Integrity
    - \* Accountability
  - Emotional Engagement
    - \* Excitement about research
    - \* Commitment to team
    - \* Trust
- **Team communication & functioning skills = learned & practiced**
  - **Idea generation**
  - **Communication**
  - **Decision making**
  - **Time management**
  - **Conflict management**

## Lessons learned from LAGOS

- Build a diverse team of experts
- Best science outcomes for individuals and team
- Build the environment and culture needed by collaboratively drafting and implementing team policies

## What are collaboratively-drafted policies?

- POLICY\* =
  - A deliberate system of principles to guide decisions and achieve rational outcomes
  - A statement of intent
  - Can include features that are: managerial, financial, administrative, scientific, and others
- \*You can call these anything you want:
  - Best practices
  - Standard Operating Procedures
  - Guidelines
  - MOUs
  - Agreements
  - Norms
  - ...

## Policies for which team activities or processes?

- Those with any of the following characteristics:
  - Follow unwritten rules or expectations
  - Involve power differentials
  - Are uncomfortable to talk about
- Personnel expectations
  - Participation/attendance
  - Workload for each person
  - Communication
  - Meeting norms
- Allocation of resources
  - Funds & other resources
  - Expectations for those receiving funds
- Authorship
  - What counts & who decides?
  - Expectations for leading manuscripts?
- Data & Code Sharing & Use
  - Internal vs external sharing
  - Mechanics of sharing
  - Credit for data & code
- Example from co-leading LAGOS:
  - 1) Authorship policy and author management styles when writing co-authored publications

## Team Authorship Policy

- *A document that describes the team's procedures and guidelines related to manuscript (MS) development*
- **What does an authorship policy provide?**
  - Best-practices for the team to strive for
  - Opportunities to discuss authorship
  - Opportunity for those with less power to contribute to decision-making
  - Clarity of expectations, roles, and responsibilities
  - Transparency in authorship decisions
  - Help in assigning credit

## What's included in an authorship policy?

1. Date or version number (living document!)
  2. Goal & guiding principles (or values) of the policy
  3. Description of the activities to warrant authorship
  4. Strategy for assigning authorship
  5. Description of article types and needs
  6. Expectations of 'lead' author(s) and 'co-authors'
- \*Best when the policy is co-created by ALL team members
  - \*Best when ALL MSs have an author-contribution statement

## Policy into practice: Manuscript management strategies

- How the **lead author(s)** manage the MS tasks, including decision-making, timelines, soliciting contributions, and delegating tasks
- What difference do these strategies make?
  - Provides clarity for co-author contributions & expectations
  - Allows for diverse range of lead author working styles
  - Facilitates alignment of manuscript type (e.g., *dissertation chapter, essays, data papers*) with best-suited strategy
  - Allows teams to balance sometimes competing goals & principles

## LAGOS guiding principles for co-authored publications

- Efficiency & productivity
- Transparency
- Accountability
- Creativity
- Inclusion & fairness
- Protection & promotion
- **Challenge:** How to balance these, sometimes competing, principles and maximize benefits for individuals and team?

## Policy into practice: Manuscript management strategies

### Outcomes of Building Research Infrastructure (Including Team Skills & Cultures)

- MANY important conversations
  - What is good science?
  - Role of power dynamics on research teams
  - Working styles
  - Conflict negotiation
  - Project, paper, & personnel management
- MANY collaborative research manuscripts
  - 40% multi-disciplinary
  - Average number of authors/paper = 8

## Articles using LAGOS-NE or LAGOS-US

- Topics by other researchers:
  - Harmful algal blooms
  - Water quality and management
  - Global change and freshwaters
  - Envir. Economics, justice, & policy
  - Modeling freshwater species
- Journals by other researchers:
  - PNAS
  - Nature Communications
  - Land Economics
  - Water Econ & Policy
  - Intelligent Systems w Applic.
  - Envir. Pollution
  - Biological Invasions
  - Diversity & Distributions

## Where to start? Team Science Resources from LAGOS

- Team policies we share
  - Authorship policy
  - Data sharing policy
  - File-sharing, including code-sharing
  - Personnel expectations
  - Values & steps toward equity
- Published papers:
  - Synergies at the intersections of team, open, & data sciences (*Cheruvilil & Soranno 2018 BioScience*)
  - Improving team functioning (*Cheruvilil et al. 2014 Front. Ecol. Environ.*)
  - Collaborative manuscript writing (*Oliver et al. 2018 Ecosphere*)
  - Authorship policy template (*Soranno & Cheruvilil 2019 FigShare*)
  - Authorship case study (*Cheruvilil et al. 2019 Online Ethics Center*)
  - Six simple steps for sharing your data (*Soranno 2019 L&O Bulletin*)

## Other Sources of Team Science Resources

- Team Science: NASEM
- Field Guide: NIH
- The Toolbox Dialogue Initiative: [tdi.msu.edu](http://tdi.msu.edu)
- A Blueprint for Creating High-Performing Teams of Ecologists and Environmental Scientists
- [scienceofteams.org](http://scienceofteams.org)
- Teams, Networks, and Networks of Networks Advancing Our Understanding and Conservation of Inland Waters

## Q & A

- Contact me: [ksc@msu.edu](mailto:ksc@msu.edu)
- Connect with me on LinkedIn: @kspencecheruvililphd
- If you want to learn more about my research:
  - [bridgelabresearch.org](http://bridgelabresearch.org)
  - [lagoslakes.org](http://lagoslakes.org)
  - [ee-stem.weebly.com/](http://ee-stem.weebly.com/)